Photosynthesis (light -> chemical energy)

Project PRISM

$6CO_2 + 6 H_2O$ — (light energy + chlorophyll) —> $C_6H_{12}O_6 + 6O_2$ (wow)
^ incredibly simplified equation; CO_2 and H_2O go through their separate processes (Calvin Cycle | PS II and I) instead of directly reacting.

General importance of photosynthesis:

Photosynthesis is the main source of organic compounds in an ecosystem.
Organisms that photosynthesise:

- Plants (main case of photosynthesis)
- Algae, cyanobacteria, protists, etc (other organisms that photosynthesize)

These organisms that photosynthesise are called **photoautotrophs**.

- ^ autotrophs (make their own food) that photosynthesise (use light) to make their own organic compounds, hence photoautotrophs.
- The end product of photosynthesis is glucose.

Heterotrophs (organisms that don't make their own food) feed primarily on autotrophs (e.g. photosynthesizing plants) to obtain nutrition.
Hence the whole ecosystem is dependent on plants, as they photosynthesise and produce the majority of glucose. These photosynthesizing plants are called producers.

Organelle that enables photosynthesis: chloroplast

- Found primarily in mesophyll layers of leaves of photosynthesising plants
- Also found in all green areas (parts of the plant that are exposed all photosynthesise)
- Comprised of a double membrane, surrounding the stroma (fluid) where part of photosynthesis takes place.
- A third membrane system inside the stroma is known as the thylakoid (sacs), arranged in stacks called grana, containing chlorophyll. The rest of photosynthesis takes place here.

How does chlorophyll aid photosynthesis?

Photosynthesising plants absorb and trap light energy to use for photosynthesis. Chlorophyll is a pigment that absorbs red and blue-violet wavelengths of light for photosynthesis, reflecting green light (hence the green parts of a plant can photosynthesise).

Factors affecting rate of photosynthesis

1. Temperature
- Enzymes involved have an optimum temperature for functioning
2. Chlorophyll concentration
- Affects amount of light energy absorbed
3. Light intensity
- Affects amount of light energy absorbed
4. Carbon dioxide concentration
- Affects the Calvin cycle
5. Presence of water
- Affects photosystems II and I
6. Surface area of leaf exposed to sunlight
- Affects amount of light energy absorbed

Stages of photosynthesis

Light dependent reactions (photosystems I and II)

Happen only in the presence of light, light is needed to provide energy for photosynthesis.

Overall function of light-dependent reactions: Convert solar energy into chemical energy in the form of NADPH and ATP.

- This chemical energy supports the light-independent reactions and fuels the assembly of sugar molecules.
- In the light-dependent reactions protein complexes and pigment molecules work together to produce NADPH and ATP from NADP+ and ADP.

The reactions place in a multiprotein complex called a **photosystem**, comprising two types embedded in the **thylakoid membrane**, photosystem II (PSII) and photosystem I (PSI)

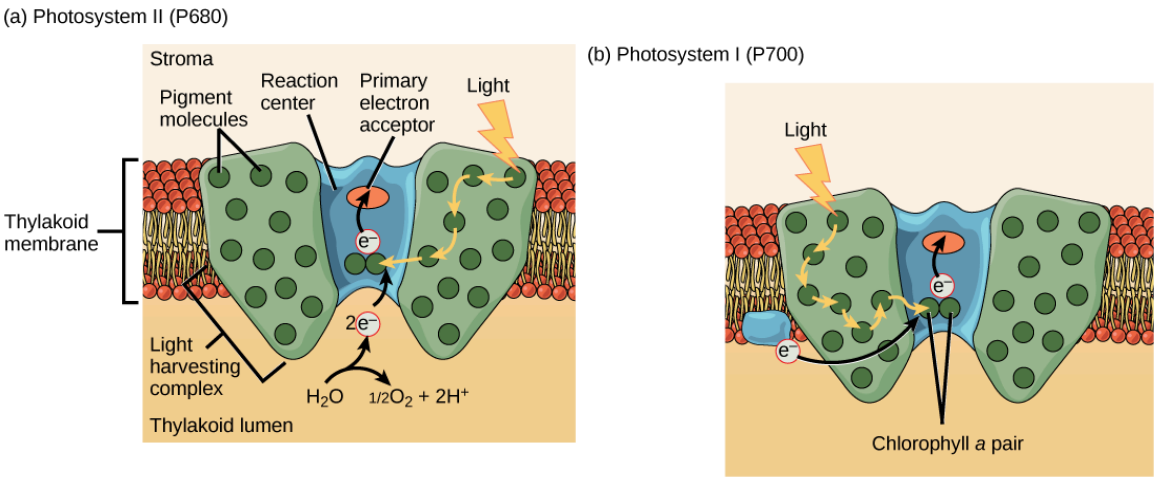
Note: PS II happens before PS I, just that PS I was discovered first, hence the naming

Basic structure of photosystems:

- A number of antenna proteins to which the chlorophyll molecules are bound surround the **reaction center** (P680 for PS II, P700 for PSI), where the photochemistry takes place.

Each photosystem is a serviced antenna complex, which **passes energy from sunlight to the reaction center**.

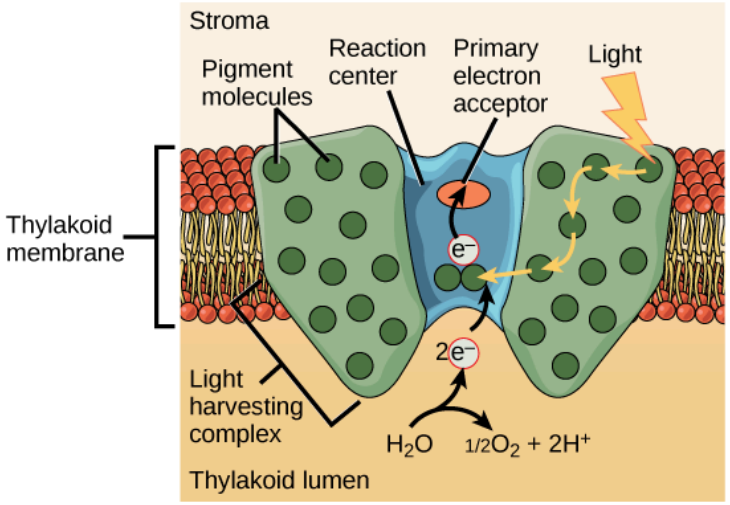
- Consists of multiple antenna proteins that contain a mixture of 300–400 chlorophyll a and b molecules as well as other pigments like carotenoids.



1. Conversion of light energy to electrons

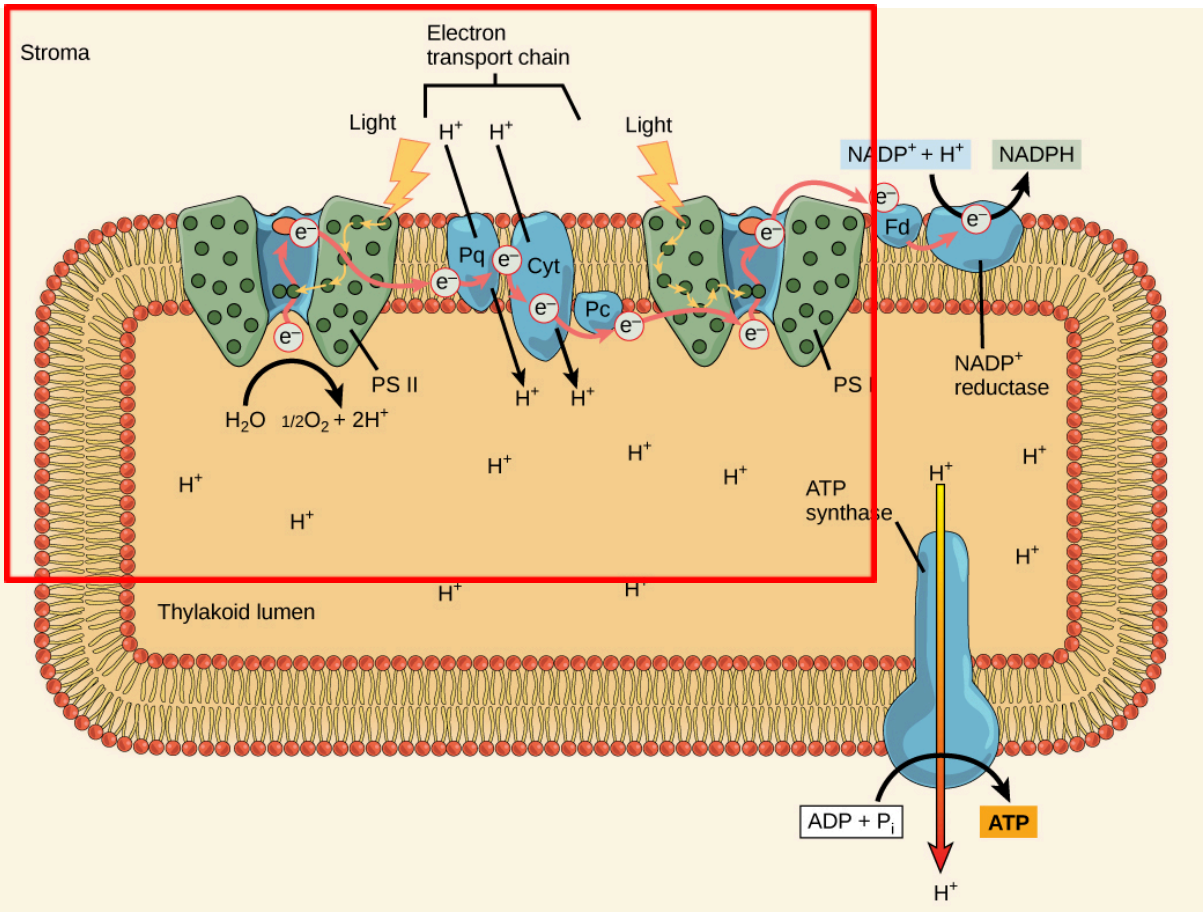
1. The absorption of a single photon, or distinct quantity or “packet” of light, by any of the chlorophylls pushes that molecule into an excited state.
2. The energy is transferred from chlorophyll to chlorophyll until it is delivered to the **reaction center**.
    - ^ Up to this point, only energy has been transferred between molecules, not electrons.
3. The reaction center contains a pair of special chlorophyll a molecules that can undergo oxidation upon excitation giving up an electron.
    - ^ It is at this step in the reaction center, this step in photosynthesis, that light energy is converted into an excited electron.

(a) Photosystem II (P680)



2. Movement of electrons down the electron transport chain

1. The reaction center of PS II delivers its high-energy electrons, one at the time, to the primary electron acceptor (pheophytin)
2. The electrons travel through the electron transport chain (Pq to cytochrome b6f complex to plastocyanine) to PS I.
    - ^ cytochrome b6f complex, an enzyme composed of two protein complexes, transfers the electrons from the carrier molecule plastoquinone (Pq) to plastocyanin (Pc).
    - cytochrome b6f also umps protons from the stroma into the thylakoid lumen using the energy from electrons moving down the electron transport chain. (creation of chemiosmotic gradient)



*pheophytin not shown in the diagram

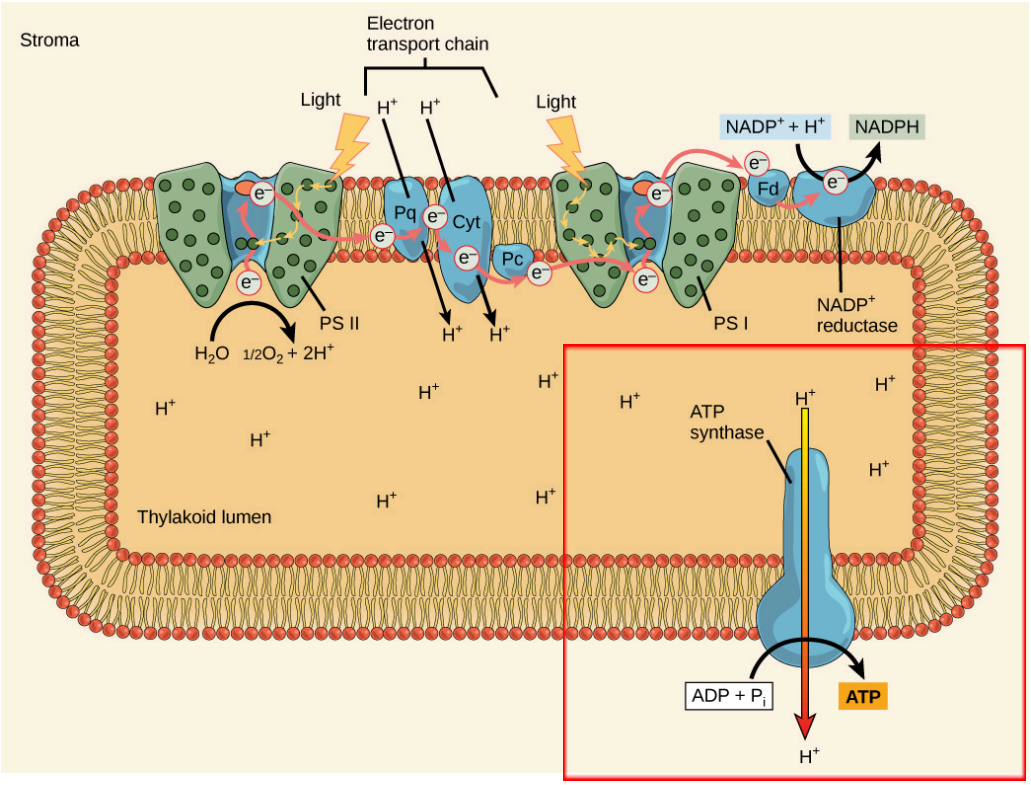
3. Photolysis of Water

1. P680's missing electron is replaced by extracting a low-energy electron from water.
2. Water is split and PSII is re-reduced.
    - Splitting one H2O molecule releases two electrons, two protons, and one atom of oxygen.
3. The protons, in addition to the ones transported by cytochrome b6f, accumulate in the thylakoid lumen and fuel photophosphorylation.
4. Splitting two molecules is required to form one molecule of diatomic O2 gas.
    - ^ 10% is used by mitochondria in the leaf to support aerobic cellular respiration.
    - The remainder escapes to the atmosphere.

4. Photophosphorylation

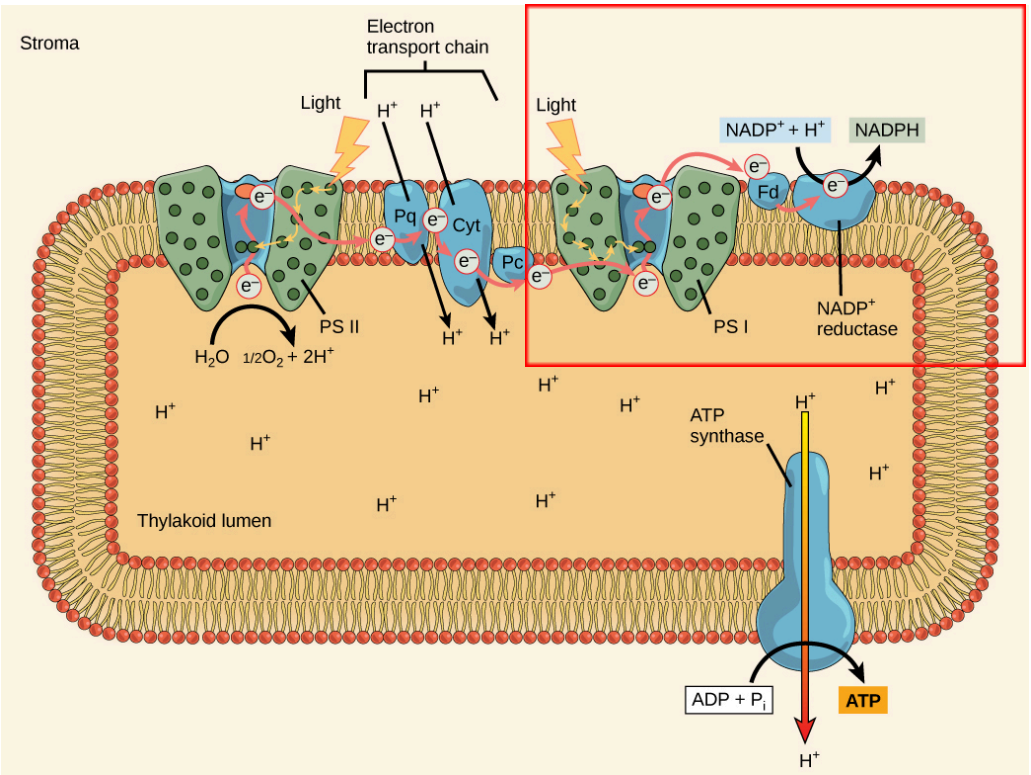
Synthesis of ATP in the light-dependent reactions -> photophosphorylation.

1. The buildup of protons inside the thylakoid lumen creates a concentration gradient.
2. The passive movement (**facilitated diffusion**) of **protons** from **high** concentration (in the <u>thylakoid lumen</u>) to **low** concentration (in the <u>stroma</u>) is harnessed to create ATP.
3. The protons build up energy at high concentrations because they all have the same electrical charge, repelling each other.
    - To release this energy, protons have to exit through an "opening".
4. Thylakoid membrane has a specialized protein channel called ATP synthase.
5. Energy from the movement of proteins allows ATP synthase to attach a third phosphate group to ADP, which forms a molecule of ATP.
    - Flow of protons through ATP synthase is called **chemiosmosis** because the ions move from an area of high to an area of low concentration across a semi-permeable membrane.



5. PS I - Reduction of NADP+

1. Electrons have lost energy prior to their arrival at PS I and must be re-energized by PS I.
2. Another photon is absorbed by the PS I antenna (similar to PS II).
3. Energy is relayed to the PSI reaction center (called P700).
4. P700 is oxidized and sends high-energy electrons through ferredoxin to the electron carrier NADP+ to form NADPH. (via NADPH reductase)



<u>PS II</u> captures the energy to create **<u>proton gradients</u>** to make **<u>ATP</u>**, and **<u>PS I</u>** captures the energy to **<u>reduce NADP+</u>** into **<u>NADPH</u>**.

- The two photosystems work together, to guarantee that the correct proportions of NADPH and ATP needed for the light-independent reactions are generated.
- Other mechanisms exist to fine tune that ratio to exactly match the chloroplast's constantly changing energy needs.

During light-dependent reactions, light energy is converted into chemical energy in the form of NADPH (reducing power) and ATP (cellular energy).

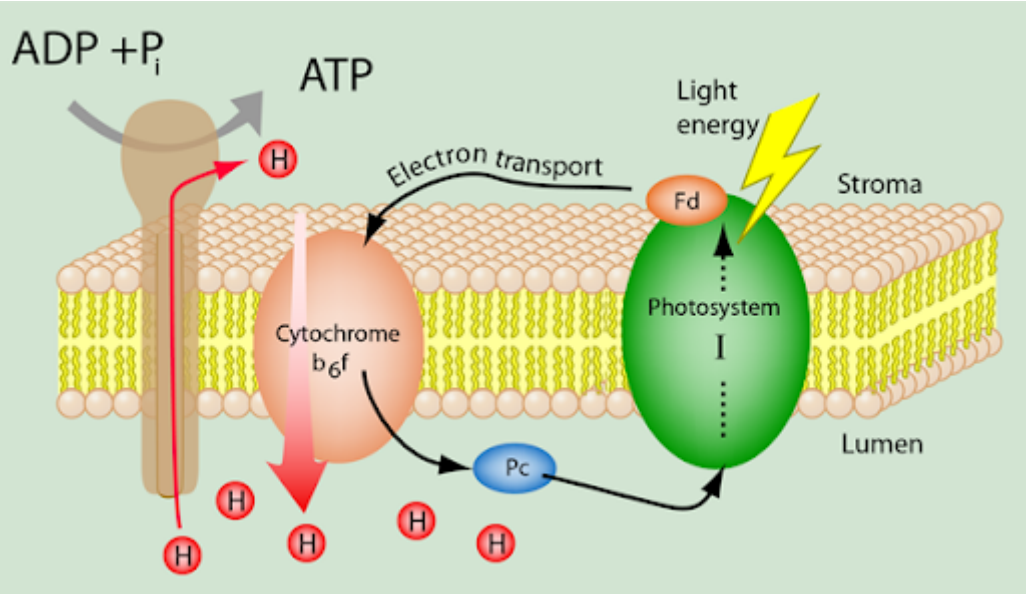
Non-Cylic Photophosphorylation

The system mentioned above involves non-cyclic photophosphorylation, where ADP is phosphorylated to ATP and NADP+ is reduced to NADPH, involving **both** PS II and I. Water acts as the electron donor while NADP+ is the final electron acceptor

Cyclic Photophosphorylation

Plants have an alternative route that can be taken by electrons when NADPH is not required.

1. Electrons, instead of passing through ferredoxin to form NADPH via NADPH reductase, enter the **cytochrome b6f complex** (responsible for pumping protons from the stroma into the thylakoid lumen)
2. This system produces more ATP and no NADPH.
3. The ability to switch between non-cyclic and cyclic photosystems allows plants to make the proper ratio of ATP and NADPH needed for the light-independent reactions.



As can be seen in the diagram, in cyclic photophosphorylation:

1. Electrons are constantly transported from PS I to cytochrome b6f.
2. The movement of electrons allows cyt b6f to continuously pump protons from the stroma into the thylakoid lumen.
3. This causes the phosphorylation of more ADP to ATP via ATP synthase.
4. Upon entering the stroma through the ETC, the electrons are once again transported to PS I to repeat the cycle.

PS I functions as the electron donor **an**d acceptor in cyclic photophosphorylation, and PS II is not involved.

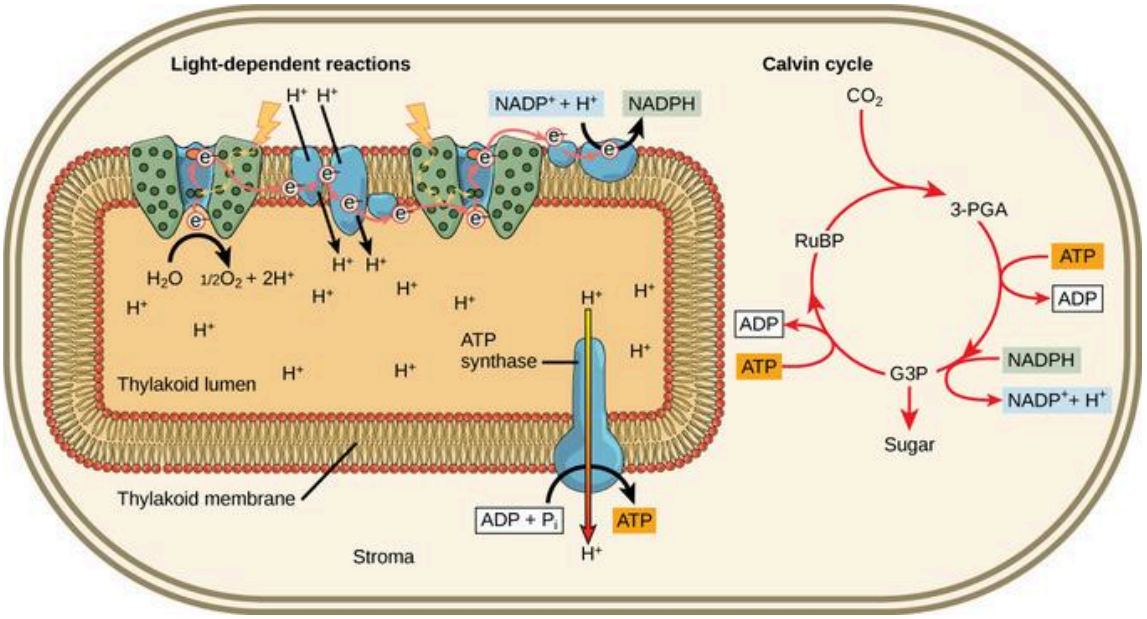
Light independent reactions (Calvin Cycle)

“Light independent” as no light energy is required, but dependent on energy from PS I and II in the form of NADPH and ATP for photosynthesis to continue.

- “Dark reactions” is the wrong term, light is still indirectly required. Calvin cycle does not happen as much in the dark

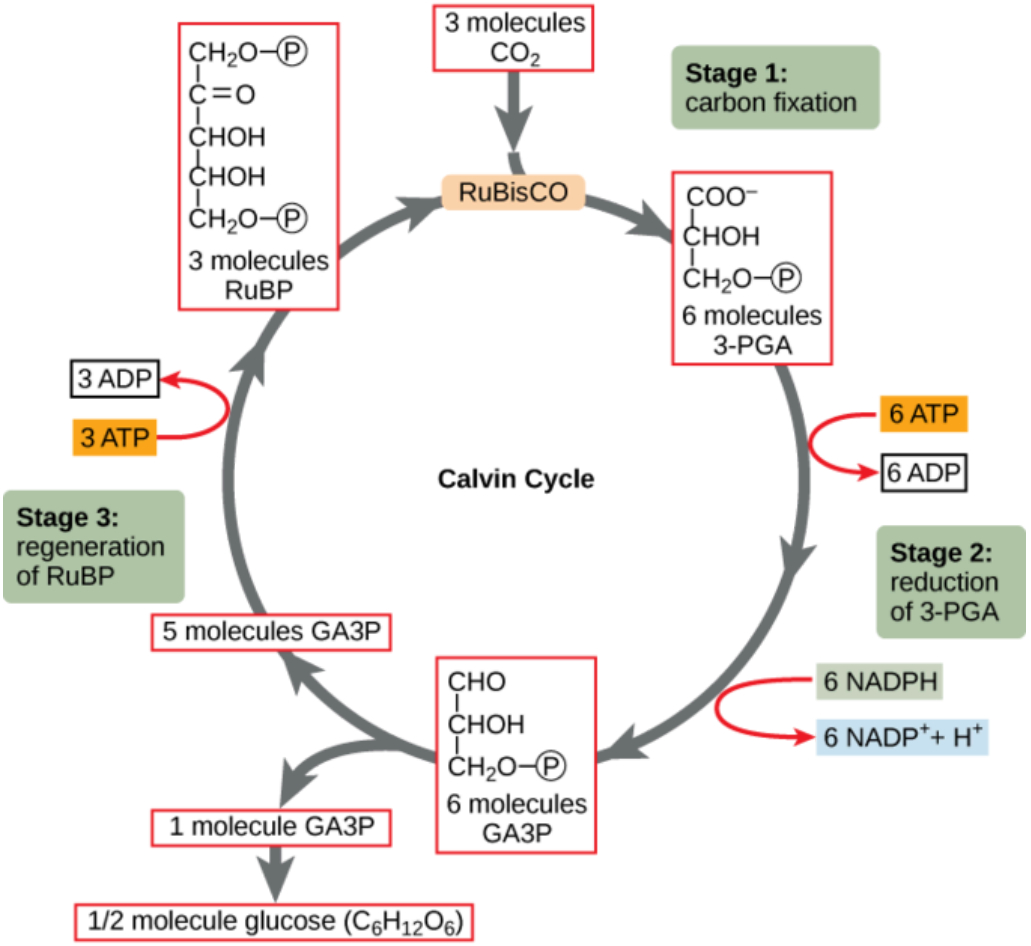
Lead up to Calvin Cycle:

- carbon dioxide (CO2) enters the leaves through stomata
- diffuses over short distances through intercellular spaces until it reaches the mesophyll cells
- CO2 diffuses into the stroma of the chloroplast, the site of light-independent reactions of photosynthesis.



Light Reactions (covered above):

Light-dependent reactions harness energy from the sun to produce chemical bonds, ATP, and NADPH. These energy-carrying molecules are made in the stroma where the Calvin cycle takes place.



The light-independent reactions of the Calvin cycle can be organized into three basic stages: fixation, reduction, and regeneration.

Stage 1: Carbon Fixation, Carboxylation of RuBP, formation of 3-PGA

1. The enzyme RuBisCO (ribulose bisphosphate carboxylase) catalyses the reaction of carbon dioxide (1C) with RuBP (ribulose bisphosphate) (5C).
2. For each CO2 molecule that reacts with one RuBP, two molecules of 3-phosphoglyceric acid (3-PGA) form.
    - ^ short-lived 6C intermediate breaks down into two 3C 3-PGA
    - 3-PGA has three carbons and one phosphate.
3. Each turn of the cycle involves only one RuBP and one carbon dioxide and forms two molecules of 3-PGA.
4. The number of carbon atoms remains the same, as the atoms move to form new bonds during the reactions
    - ^ 3 atoms from 3CO2 + 15 atoms from 3RuBP = 18 atoms in 3 atoms of 3-PGA

This process is called carbon fixation because CO2 is “fixed” from an inorganic form into organic molecules.

Stage 2: Reduction of 6 3-PGA to 6 GA3P (glyceraldehyde-3-phosphate)

1. ATP and NADPH (from PS I and II) are used to convert the six molecules of 3-PGA into six molecules of a chemical called glyceraldehyde 3-phosphate (G3P).
2. 3-PGA is reduced using electrons supplied by NADPH to GA3P.
    - ^ reduction is the gain of an electron by an atom or molecule.
3. Six molecules of both ATP and NADPH are used. ATP is converted back into ADP as the terminal phosphate group is lost. NADPH converts back to NADP+ as the electron and hydrogen atom are lost.
    - ^ both of these molecules return to PS I and II to be reused and reenergized.

Stage 3: Regeneration of RuBP

1. One of the 6 generated GA3P molecules leaves the calvin cycle and is sent to the cytoplasm to form other compounds (glucose, etc)
2. The cycle must be completed three times to produce a single three-carbon GA3P molecule, and six times to produce a six-carbon glucose molecule.
    - RuBP: 5C compound
    - GA3P: 3C compound
    - 1 “turn” of Calvin cycle: 2 G3P (6C) formed, 1 RuBP (5C) used
    - 3 “turns” of Calvin cycle: 6 GA3P (18C) formed, 3 RuBP (15C) used
    - Hence, 5 GA3P used to regenerate 3 RuBP, one net GA3P released.
    - All the RuBP is regenerated and cycle continues
3. Three more molecules of ATP are used in the regeneration reactions.

Key Points

- The Calvin cycle refers to the light-independent reactions in photosynthesis that take place in three key steps.
- Although the Calvin Cycle is not directly dependent on light, it is indirectly dependent on light since the necessary energy carriers (ATP and NADPH) are products of light-dependent reactions.
- In fixation, the first stage of the Calvin cycle, light-independent reactions are initiated; CO2 is fixed from an inorganic to an organic molecule.
- In the second stage, ATP and NADPH are used to reduce 3-PGA into G3P; then ATP and NADPH are converted to ADP and NADP+, respectively.
- In the last stage of the Calvin Cycle, RuBP is regenerated, which enables the system to prepare for more CO2 to be fixed.

Key Terms

- **light-independent reaction**: chemical reactions during photosynthesis that convert carbon dioxide and other compounds into glucose, taking place in the stroma
- **rubisco**: (ribulose bisphosphate carboxylase) a plant enzyme which catalyzes the fixing of atmospheric carbon dioxide during photosynthesis by catalyzing the reaction between carbon dioxide and RuBP
- **ribulose bisphosphate**: an organic substance that is involved in photosynthesis, reacts with carbon dioxide to form 3-PGA

C4 and Crassulacean Acid Metabolism (CAM) Plants

The aforementioned Calvin cycle relies on C3 photosynthesis. However, C4 and CAM plants have evolved unique adaptations that allow them to thrive in specific environmental conditions, particularly those involving heat and aridity.

Rubisco needs a high CO2 environment to function efficiently.

- If the ratio of oxygen to CO2 gets too high, rubisco will bind to oxygen instead and waste energy in the process.
- This is called **photorespiration** and accounts for a large amount of yield loss for crops in hot areas.

Reasons for photorespiration

- If it is too hot or dry, plants often close their stomata to prevent water loss.
- This prevents CO2 from entering the leaf, as well as preventing O2 from exiting.
- Oxygen builds up inside the leaf and photorespiration happens instead of the Calvin cycle.
- Though this wastes energy for the plant, preventing water loss is often a larger priority.

Some plants have evolved special ways of performing photosynthesis that prevent or limit photorespiration.

C4 Photosynthesis

<u>C4 Photosynthesis is used by plants adapted to hot environments.</u>

- In most photosynthesis, the first product of the Calvin cycle is a 3 carbon compound, so this type of photosynthesis is called C3.
- For plants adapted to particularly hot environments, the first compound formed has 4 carbon atoms, hence C4 photosynthesis.
- This process is facilitated by the enzyme phosphoenolpyruvate (PEP) carboxylase
    - It has a higher affinity for CO2 and does not react with oxygen, thus minimizing photorespiration.

<u>In C4 plants, carbon fixation happens in the mesophyll cells and the Calvin cycle happens in the bundle sheath cells.</u>

- Rubisco is restricted to the bundle sheath cells of the leaf.
- Carbon dioxide is converted into an acid and transported into the bundle sheath cells where it will be converted back into CO2.
- Keeps CO2 concentration high where rubisco is active, <u>preventing photorespiration</u>.

<u>C4 photosynthesis is particularly advantageous in hot, dry environments</u>

- Allows plants to minimize water loss by partially closing their stomata while still efficiently fixing CO2.
- This avoids the energy losses associated with photorespiration.
- Common examples of C4 plants include <u>maize, sugarcane, and sorghum</u>, which are typically found in tropical and subtropical regions where they can outcompete C3 plants under high light intensity and temperature conditions.

CAM Photosynthesis

CAM Photosynthesis is used by plants adapted to dry environments (often desert environments).

- It is too hot and/or dry to keep stomata open during the day, so they only open them at night.
- However, there is no light at night to do photosynthesis.

To solve this, CAM plants have evolved to take in CO_2 at night and store it in the central vacuole in the form of an acid.

- During the day, the acid is converted back into CO_2, and the Calvin cycle can take place alongside the electron transport chain.

In both CAM and C4 photosynthesis, compounds must be formed, transported, and broken back apart again. Each of these tasks costs energy to perform, but it outweighs the energy lost by photorespiration.

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